

Smart Home AIoT – Test Data Collection Sheet – Phase H – AI Agent (LINE Thai + Alexa EN + Deep Reasoning)

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- Instructions for use:
1. Run each test 10 times consecutively (or across sessions, as practical)
 2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
 3. Pass criteria: Actual matches Expected description
 4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
 5. Use Comments for any anomalies or special observations

- Statistical formulas:
- Pass Rate = (Pass Count / 10) × 100%
 - Avg RT = (response times) / 10
 - Variance = (xi -)² / n
 - Std Dev = √variance

Coverage Matrix

Phase	Description	Count
Phase H	Phase H – AI Agent (LINE Thai + Alexa EN + Deep Reasoning)	39
	TOTAL	39

Phase H – AI Agent (LINE Thai + Alexa EN + Deep Reasoning)

H01	LINE: "เปิดไฟห้องนอน"				Phase H		
Method: สั่ง LINE							
Expected: switch.bedroom_light ON + Thai reply							
#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10		Pass Rate: ____%		Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: ส่ง LINE

Expected: all switches OFF

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

H03

LINE: "นอนแล้ว"

Phase H

Method: ส่ง LINE

Expected: Sleep mode + cascade

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H04

LINE: "ออกจากบ้าน"

Phase H

Method: ส่ง LINE

Expected: Away mode + cascade

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H05

LINE: "อุณหภูมิครัว"

Phase H

Method: ส่ง LINE

Expected: sensor reply with °C value

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H06

LINE: "เปลี่ยนสีไฟเป็นแดง"

Phase H

Method: สั่ง LINE

Expected: Tuya bulb red

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H07

LINE: "ปาร์ตี้!"

Phase H

Method: ส่ง LINE

Expected: Party scene (colorloop) fires

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H08

LINE: "มีคนบุก"

Phase H

Method: ส่ง LINE

Expected: Security mode + intrusion response

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H09

LINE: "แก๊สรั่ว"

Phase H

Method: ส่ง LINE

Expected: Gas alert simulated response

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: both actions fire

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H11	Alexa: "turn on bedroom light"	Phase H
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Method: พูด Alexa

Expected: switch.bedroom_light ON + TTS confirm

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10 Pass Rate: ____% Avg RT: ____ ms Std Dev: ____ ms

Comments:

H12

Alexa: "kitchen fan on"

Phase H

Method: พูด Alexa

Expected: fan ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: cover.garage_door open

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: Away cascade

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: TTS reply with °C

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: Tuya colorloop

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: lock + servo หมุน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H18

Alexa: "tell home control I am going to sleep"

Phase H

Method: พูด Alexa

Expected: Sleep mode via AI Agent

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H19

Alexa: "tell home control what model is the living room bulb"

Phase H

Method: พูด Alexa

Expected: TTS "Tuya TS0505B"

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: Party scene via AI Agent (Thai)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: เช็ค gas + เต้า + suggest ไฟครัว

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: สั่ง LINE

Expected: scan gas ยืนยันปลอดภัย

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: list device ON propose ปิด

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H24

LINE Reasoning: "ออกจากบ้าน ลืมอะไรมัย"

Phase H

Method: ส่ง LINE

Expected: checklist เตรียม Away

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: Sleep mode + suggest Sunrise scene

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: อ่าน temp + ปิดพัดลม + Tuya warm

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: check device_tracker + GPS + door history

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง LINE

Expected: time-aware Sleep mode suggestion

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: gas check + stove + suggest light

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: bedroom temp + adjust

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: smart triage

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: multi-sensor scan

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: multi-action sleep prep

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: weather + temp + Sunrise scene

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: pre-Away checklist

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: พูด Alexa

Expected: time + state inference

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง 2 turns

Expected: follow-through action

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: ส่ง 2 turns

Expected: mode/scene back to normal

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

H39

Compound Alexa: "I am leaving home but feed the dog first"

Phase H

Method: พูด Alexa

Expected: break down: feed + Away

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments: